A Certified Nonconvex Wrapping of a Disk with Volume Greater than the Sphere

Sho Hoshino

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Abstract

We give an explicit computer-assisted construction of a nonconvex solid wrapped by a disk without stretching. For every $R > 0$ we construct a continuous 1-Lipschitz map from the closed disk $D_R \subset \mathbb{R}^2$ onto an embedded polyhedral 2-sphere bounding a nonconvex solid B_R of volume

$$\mathrm{Vol}(B_R) = CR^3, \qquad C = 0.17984349416291615\ldots.$$

The constant C is given by an exact rational number. In particular, for $R = \pi$ the volume exceeds $4\pi/3$, the volume of the unit ball. Under the 1-Lipschitz formalization of wrapping used by Nandakumar, this gives a small explicit and exactly certified witness for his Question 1. We do not claim priority for abstract existence alone: Nandakumar explicitly points to the volume-increasing isometric deformations of Bleecker and Pak as potentially implying the conjectural phenomenon [4, 5, 1]. Our contribution is a direct 17-vertex, 30-face construction with an exact, independently reproducible certificate and a quantitative lower bound. The requirement that the comparison body be nonconvex remains essential to the intent of Question 1, and our result should not be read as proving Nandakumar's full conjecture for the disk.

For context, the classical Mylar balloon is the sharp volume maximizer among convex bodies of revolution of intrinsic diameter R, with volume $0.152315527\ldots R^3$. We give below an explicit radial 1-Lipschitz map from D_R onto this surface, making its wrappability by the single disk completely self-contained. Thus the certified nonconvex example is about 18.07% larger than this sharp axisymmetric convex benchmark as well as larger than the sphere. The construction is based on a regular decagon with a single pleat in its boundary identification and is verified by two independent exact-arithmetic programs. The certificate checks strict nonexpansion on all 30 source triangles, the topology of the quotient complex, all 435 triangle-pair intersections, a nonconvexity witness, and the exact enclosed volume. No floating-point assertion is used in the proof certificate.

1 Introduction

Nandakumar recently formulated a volume-maximization problem for wrapping three-dimensional solids by a planar sheet that may be folded or wrinkled but not stretched or torn [1]. In his formalization, a planar region D wraps a compact connected solid when the relevant closed surface is realized by a 1-Lipschitz image of D, with the boundary of D glued to itself in the image. His main conjecture is explicitly *nonconvex*: for every convex solid B wrappable by a given nondegenerate connected sheet D, there should exist a nonconvex solid B' wrappable by the same sheet with

$$\mathrm{Vol}(B') > \mathrm{Vol}(B).$$

As a concrete test case he takes D to be the disk of radius π and B to be the unit ball, and asks whether that disk can wrap a *nonconvex* solid of volume strictly larger than $4\pi/3$ [1, Claim 1 and

Question 1]. Thus the adjective “nonconvex” is not incidental: producing a larger convex body would not answer the stated subquestion.

There is substantial prior evidence for the underlying existence phenomenon. Bleecker proved that the surface of a generic convex polyhedron admits an isometric deformation enclosing greater volume [4]; Pak later gave a particularly simple explicit inflation of the cube and summarized the broader polyhedral picture [5]. Nandakumar himself therefore notes that his conjecture may conceivably be a corollary of this circle of results [1, Section 3]. We accordingly make no claim that abstract existence for Question 1 is new. Establishing a complete indirect implication in the present disk setting would require additional approximation and wrappability arguments that are unnecessary for our theorem, so we do not assert such an implication here.

The purpose of this note is instead to give a very small, fully explicit, quantitatively strong witness. We specify the source triangulation, the target coordinates, and the piecewise-affine map completely. The proof does not require an existence theorem for an unspecified folding. The only computer-assisted part is the exhaustive verification of a finite collection of exact algebraic inequalities and intersection tests. Two implementations using different arithmetic and different triangle-intersection algorithms are supplied as supplementary material.

It is useful to separate this statement from the stronger main conjecture. Our theorem treats the particular convex comparison body chosen in Question 1, namely the unit ball. It does *not* show that every convex body wrappable by the disk is beaten by our example, nor that our example exceeds the supremum of volumes of convex bodies wrappable by the disk. In particular, the sphere is not the strongest known convex benchmark; the classical Mylar balloon already gives a larger convex body wrappable by the same disk after matching intrinsic diameter. We record this comparison precisely in section 2.

The geometry was discovered by numerical search, but the theorem below depends only on the final rational coordinate certificate. The construction can be viewed as a “one-pleat” analogue of a dumpling: most of the boundary forms a doubled seam while one short branch stores excess boundary length. The informal analogy plays no role in the proof.

Theorem 1 (Certified one-pleat disk wrapping)**.** *Let $D_R \subset \mathbb{R}^2$ be the closed disk of radius $R > 0$. There exists a continuous 1-Lipschitz map*

$$F_R : D_R \longrightarrow \mathbb{R}^3$$

whose image is an embedded nonconvex polyhedral 2-sphere ∂B_R. The bounded solid B_R has volume

$$\mathrm{Vol}(B_R) = CR^3, \tag{1}$$

where

$$C = \frac{359686988325832312742347877865122384142690317017149}{2000000000000000000000000000000000000000000000000000} = 0.17984349416291615\ldots.$$

In particular, $C > 0.179843$.

Corollary 2. *A disk of radius π wraps a nonconvex solid of volume strictly greater than $4\pi/3$.*

Indeed, Theorem 1 gives $\mathrm{Vol}(B_\pi) > 0.179843\pi^3$. Since $\pi > 3$,

$$0.179843\pi^3 > 9(0.179843)\pi = 1.618587\pi > \frac{4}{3}\pi.$$

Thus Corollary 2 provides an explicit exactly certified witness for Question 1 of [1] under its stated 1-Lipschitz wrapping formalization.

2 Relation to Nandakumar's conjecture and a convex benchmark

The logical role of Corollary 2 is worth making explicit. Nandakumar's Claim 1 compares *each* wrappable convex body with some larger wrappable nonconvex body [1, Claim 1]. Question 1 fixes one such convex body, the unit ball, and asks for a corresponding nonconvex improvement [1, Question 1]. Our construction supplies a particularly small explicit improvement, but we do not claim that the existence of some improvement is new: the volume-increasing polyhedral deformations of Bleecker and Pak provide mechanisms closely related to this question [4, 5], as Nandakumar observes [1, Section 3]. The novelty sought here is the concrete certified witness and its quantitative lower bound. In particular, the construction does not prove Claim 1 for all convex bodies wrappable by a disk.

There is also a natural convex comparison stronger than the sphere. The classical Mylar balloon is the convex surface of revolution arising from the fixed-meridian-length volume-maximization problem studied by Paulsen and subsequently by Mladenov and Oprea [2, 3]. For a convex body of revolution, the intrinsic diameter equals the north–south meridian length R. Indeed, in meridional arclength coordinates (s, θ) the surface metric has the form $ds^2 + \varrho(s)^2 d\theta^2$, with $0 \le s \le R$. Any two points at arclengths s_1, s_2 can be joined through one of the poles with length at most

$$\min\{s_1 + s_2,\, 2R - s_1 - s_2\} \le R,$$

whereas every path from the north pole to the south pole has length at least $\int |ds| \ge R$. Thus fixing intrinsic diameter is equivalent, in this convex axisymmetric class, to fixing the generating meridian length.

For completeness, the sharpness of the Mylar profile in this class admits a short global argument. Parametrize the meridian by arclength and write it as $(\varrho(s), z(s))$. For a general convex meridian these coordinate functions are Lipschitz; throughout this paragraph derivatives are understood almost everywhere. Convexity makes z monotone and ϱ increase from 0 to its maximum a and then decrease to 0. Hence

$$\frac{V}{\pi} = \int_0^R \varrho^2 \sqrt{1 - \varrho'^2}\, ds.$$

For every $0 \le \varrho \le a$, Cauchy–Schwarz gives

$$\varrho^2 \sqrt{1 - \varrho'^2} + |\varrho'| \sqrt{a^4 - \varrho^4} \le a^2.$$

After integration and the monotonicity of the two halves of the meridian,

$$\frac{V}{\pi} \le a^2 R - 2 \int_0^a \sqrt{a^4 - u^4}\, du = a^2 R - 2Ka^3, \qquad K := \int_0^1 \sqrt{1 - u^4}\, du.$$

The right-hand side is maximized at $a = R/(3K)$. Since $K = 2I/3$ for the constant I below, this is $a = R/(2I)$. Equality in Cauchy–Schwarz requires, on each monotone half of the meridian,

$$|\varrho'| = \sqrt{1 - (\varrho/a)^4}, \qquad \sqrt{1 - \varrho'^2} = \frac{\varrho^2}{a^2}.$$

Integrating the first relation from a pole to the equator and back gives

$$R = 2a \int_0^1 \frac{du}{\sqrt{1 - u^4}} = 2aI,$$

so the equality profile has exactly the maximizing value $a = R/(2I)$. This is the classical Mylar profile, and hence the upper bound is attained. Thus the Mylar balloon is the sharp volume maximizer among convex bodies of revolution with intrinsic diameter R.

Let

$$I = \int_0^1 \frac{du}{\sqrt{1-u^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\Gamma(3/4)}.$$

With intrinsic diameter R, its equatorial radius is $r = R/(2I)$ and its volume is

$$V_{\text{Mylar}}(R) = \frac{\pi}{12I^2}R^3 = \frac{4}{3}\left(\frac{\Gamma(3/4)}{\Gamma(1/4)}\right)^2 R^3 = 0.152315527\ldots R^3. \tag{2}$$

The wrappability of this comparison body by the *single* disk D_R can be checked directly. Parametrize a generating meridian by arclength $s \in [0, R]$ and write it as $(\varrho(s), z(s))$, with the north and south poles at $s = 0$ and $s = R$. Thus

$$\varrho(0) = \varrho(R) = 0, \qquad \varrho'(s)^2 + z'(s)^2 = 1.$$

In polar coordinates (s, θ) on D_R, define

$$F(s,\theta) = \big(\varrho(s)\cos\theta,\ \varrho(s)\sin\theta,\ z(s)\big).$$

Away from the poles, the pullback of the surface metric is

$$F^*g = ds^2 + \varrho(s)^2 d\theta^2.$$

Since $|\varrho'| \le 1$ and $\varrho(0) = 0$, one has $0 \le \varrho(s) \le s$ for all $s \in [0, R]$. Hence

$$F^*g \le ds^2 + s^2 d\theta^2,$$

the Euclidean polar metric on D_R. It follows by integrating lengths of curves (and then by continuity at the poles) that F is 1-Lipschitz. The map is onto, and the entire boundary circle $s = R$ is sent to the south pole. Thus D_R wraps the Mylar body in Nandakumar's 1-Lipschitz sense. Notice that this uses the Mylar balloon only as a target convex body; it does not identify the source sheet D_R with the doubled circular sheets used in the classical physical Mylar construction.

For comparison, the same direct radial construction wraps the sphere of radius R/π: its parallel radius at meridional distance s from the north pole is $(R/\pi)\sin(\pi s/R) \le s$. Its volume is therefore

$$V_{\text{sph}}(R) = \frac{4}{3\pi^2}R^3 = 0.1350949115\ldots R^3.$$

Thus the three explicit benchmarks are

body	volume divided by R^3
sphere	$0.1350949115\ldots$
sharp convex body-of-revolution Mylar benchmark	$0.1523155270\ldots$
certified nonconvex body of this paper	$0.1798434942\ldots$

Our certified example therefore exceeds the Mylar benchmark by about 18.07%. This strengthens the quantitative comparison, but the theorem's relevance to Nandakumar's Question 1 comes from the combination of *nonconvexity* and volume exceeding the unit ball, not from any assertion that the unit ball is convex-optimal.

3 The source complex and the one-pleat seam

It suffices to construct the case $R = 1$; scaling source and target by R then multiplies volume by R^3.

Let H be the regular decagon inscribed in the unit circle, with outer vertices

$$o_k = (\cos(2\pi k/10), \sin(2\pi k/10)), \qquad k = 0, \ldots, 9.$$

Set

$$\rho = \frac{13}{20}, \qquad i_k = \rho o_k,$$

and let $z = (0, 0)$ be the center. Triangulate H by the 30 triangles

$$\begin{aligned}&(z, i_k, i_{k+1}),\\ &(i_k, o_k, o_{k+1}),\\ &(i_k, o_{k+1}, i_{k+1}),\end{aligned}$$

with indices modulo 10. All squared source lengths lie in $\mathbb{Q}(\sqrt{5})$.

The ten outer vertices are identified in cyclic order as

$$A_2,\ A_1,\ J,\ C,\ J,\ B_1,\ B_2,\ B_1,\ J,\ A_1. \tag{3}$$

Thus the boundary traces the graph

$$A_2 - A_1 - J - C - J - B_1 - B_2 - B_1 - J - A_1 - A_2.$$

Every seam edge is traversed twice in opposite directions, and the branch $J - C - J$ is the single pleat. The ten inner vertices and the center remain distinct. Hence the quotient has 17 vertices.

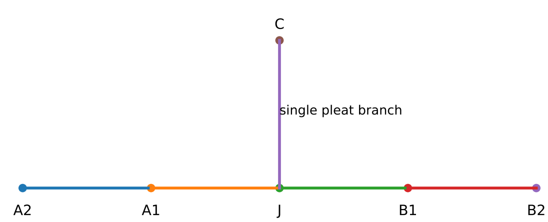

Figure 2: Combinatorial seam graph. The branch JC is the single pleat. The actual target coordinates are not coplanar and are listed in section 4.

4 Explicit target coordinates

Let

$$\lambda = \frac{99999}{100000}.$$

Each target vertex is λ times the finite-decimal vector in table 1. Every displayed decimal is interpreted as an exact rational number. The affine image of each source triangle is determined by the images of its three vertices.

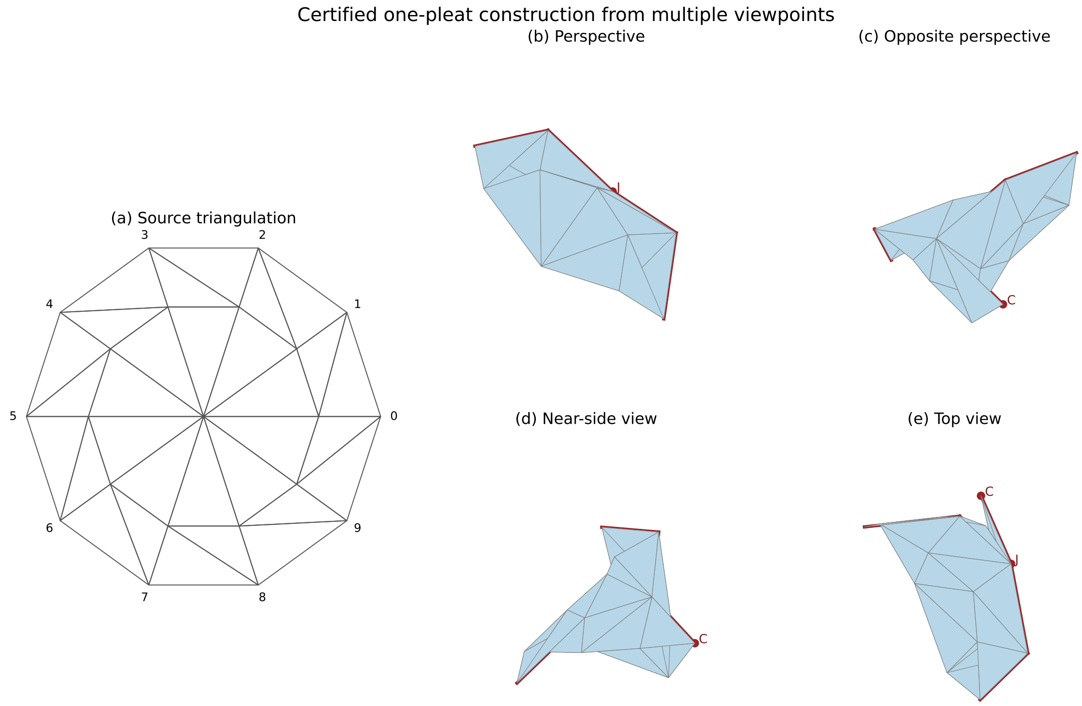

Figure 1: The source triangulation and four opaque views of the certified target surface. Red edges indicate the seam graph; the points C and J are marked in the target views. In particular, the opposite, near-side, and top views make clear that C is a vertex of the polyhedral surface, not an interior point. The drawings are for visualization only; the proof uses the exact coordinates in section 4.

Table 1: Base target coordinates before multiplication by $\lambda = 99999/100000$.

vertex	x	y	z
J	0.114996453431	0.652042463578	0.359274110539
A_1	0.713527595885	0.556758865942	0.238247176413
A_2	0.904510611077	0.150919970833	-0.186942534628
B_1	-0.269259146678	0.427202794161	0.787947375499
B_2	-0.411232941130	-0.173302559313	0.822628398965
C	-0.407447365206	0.632905197995	0.029649729336
z	0	0	0
I_0	0.650000000000	0	0
I_1	0.525861046344	0.382060413990	0
I_2	0.201227167335	0.617682813279	0.021807551587
I_3	-0.138361314859	0.605243587298	-0.192448295874
I_4	-0.192247060016	0.586039489637	0.205179883277
I_5	-0.332154604650	0.312020954927	0.463482731389
I_6	-0.424437564120	-0.078307526511	0.486025395896
I_7	-0.126467920004	0.156558885543	0.618057586773
I_8	0.208351084601	0.348929212409	0.507285156764
I_9	0.525861046344	0.263795278601	0.276373318043

Let $g : H \to \mathbb{R}^3$ denote the resulting continuous piecewise-affine map. Continuity holds because adjacent triangles use the same target coordinates on their common vertices, including the identifications in (3).

5 Exact certification

We now establish the geometric properties of g. All sign and intersection assertions in this section are checked by exact arithmetic in the supplied programs; the verification methods are described explicitly so that the certificate is finite and reproducible.

5.1 Strict nonexpansion on every triangle

For a source triangle choose two edge vectors u, v from a common vertex and set

$$S = \begin{pmatrix} \langle u, u\rangle & \langle u, v\rangle \\ \langle u, v\rangle & \langle v, v\rangle \end{pmatrix}.$$

Let T be the corresponding Gram matrix of the target edge vectors. A tangent vector with coordinate column $\xi \in \mathbb{R}^2$ has squared source length $\xi^T S\xi$ and squared target length $\xi^T T\xi$. Thus the affine triangle map is strictly nonexpansive if

$$S - T \succ 0.$$

Lemma 3. *For each of the* 30 *source triangles, $S - T$ is positive definite. Hence g is* 1*-Lipschitz on H.*

Proof. For a real symmetric 2×2 matrix, positive definiteness is equivalent to positivity of the two diagonal entries and of the determinant. The verifier computes, for each triangle,

$$(S - T)_{11} > 0, \qquad (S - T)_{22} > 0, \qquad \det(S - T) > 0.$$

The source terms lie in $\mathbb{Q}(\sqrt{5})$ and the target terms are rational; signs in $\mathbb{Q}(\sqrt{5})$ are reduced to rational comparisons by squaring only after the signs of the rational coefficients are known. Thus no floating-point sign decision is used.

To pass from triangles to the whole decagon, take $x, y \in H$. Convexity of H implies that the segment $[x, y]$ remains in H and crosses finitely many source triangles. On each subsegment the image curve has length at most the source subsegment length. Summing yields

$$|g(x) - g(y)| \le \mathrm{length}(g([x, y])) \le |x - y|.$$

□

5.2 The quotient is a sphere

Lemma 4. *The quotient triangulation is a connected closed orientable triangulated* 2*-manifold with*

$$V = 17, \qquad E = 45, \qquad F = 30,$$

and hence is topologically S^2.

Proof. The exact combinatorial verifier checks that every edge belongs to exactly two faces, every vertex link is one cycle, the complex is connected, and the listed face orientations agree oppositely along each shared edge. Hence it is a connected closed orientable 2-manifold. Its Euler characteristic is

$$17 - 45 + 30 = 2,$$

so the classification of closed orientable surfaces gives S^2. □

5.3 No unintended triangle intersections

Lemma 5. *The geometric realization of the quotient complex is embedded in* $\mathbb{R}^3$.

Proof. There are $\binom{30}{2} = 435$ unordered face pairs. The independent verifier tests intersection by solving exactly

$$A_0 + \alpha(A_1 - A_0) + \beta(A_2 - A_0) = B_0 + \gamma(B_1 - B_0) + \delta(B_2 - B_0)$$

under the barycentric inequalities

$$\alpha, \beta, 1 - \alpha - \beta, \gamma, \delta, 1 - \gamma - \delta \geq 0.$$

The three coordinate equations have rank 3 in every tested pair, leaving one affine parameter. The six inequalities become rational interval constraints on that parameter.

The exhaustive outcome is:

prescribed common simplex	number of pairs	exact intersection
none	260	empty
one vertex	130	that vertex only
one edge	45	that edge only

Thus there are no unintended self-intersections. Together with Lemma 4, the realization is an embedded PL 2-sphere. By Jordan–Brouwer separation it bounds a compact solid, denoted B_1. ☐

5.4 Nonconvexity

Lemma 6. *The bounded solid* B_1 *is nonconvex.*

Proof. Consider the face (z, I_0, I_1) and let n be its oriented normal. Exact rational arithmetic gives

$$n \cdot (J - z) = \frac{178438386772277185863216936460667247207}{2000000000000000000000000000000000000000} > 0,$$

whereas

$$n \cdot (A_2 - z) = -\frac{23211889835379489551564573316040740141}{500000000000000000000000000000000000000} < 0.$$

Thus vertices of the boundary lie on both sides of the plane of a boundary triangle. Every face plane of a convex polyhedron is supporting, so this is impossible for a convex solid. ☐

5.5 Exact volume

Lemma 7. *The volume enclosed by the target sphere equals the constant* C *in* (1).

Proof. The face orientations are globally consistent. Therefore the oriented tetrahedral formula gives

$$\mathrm{Vol}(B_1) = \frac{1}{6}\left|\sum_{(a,b,c)\in\mathcal{F}} \det(a, b, c)\right|.$$

All target coordinates are rational. Exact evaluation yields

$$\mathrm{Vol}(B_1) = \frac{359686988325832312742347877865122384142690317017149}{2000000000000000000000000000000000000000000000000000},$$

which is $0.17984349416291615\ldots > 0.179843$. ☐

6 From the decagon to the disk

The preceding construction uses only the regular decagon H, but the statement concerns the entire disk. The transfer is immediate from metric projection.

Lemma 8. *Let $\pi_H : D_1 \to H$ be Euclidean nearest-point projection. Then π_H is 1-Lipschitz and onto. Consequently*

$$F_1 := g \circ \pi_H : D_1 \to \mathbb{R}^3$$

is 1-Lipschitz and has image ∂B_1.

Proof. For $x, y \in D_1$, set $p = \pi_H(x)$ and $q = \pi_H(y)$. The standard projection inequalities for a closed convex set are

$$(x-p)\cdot(q-p) \le 0, \qquad (y-q)\cdot(p-q) \le 0.$$

Adding gives

$$|p-q|^2 \le (p-q)\cdot(x-y) \le |p-q|\,|x-y|,$$

and hence $|p-q| \le |x-y|$. The projection fixes every point of H, so it is onto. Therefore F_1 is 1-Lipschitz and

$$F_1(D_1) = g(H) = \partial B_1.$$

□

Lemma 9 (Boundary self-gluing)**.** *The restriction $\pi_H|_{\partial D_1} : \partial D_1 \to \partial H$ is onto. Under $F_1 = g \circ \pi_H$, every relative interior point of a seam edge has at least two distinct preimages on ∂D_1, coming from the two oppositely oriented boundary edges in* (3)*. Thus no boundary arc remains unglued.*

Proof. Let p lie in the relative interior of an edge e of the inscribed decagon H, and let n_e be its outward unit normal. Because p lies strictly inside the unit circle, the ray $p + tn_e$, $t \ge 0$, meets ∂D_1 at some point x. Since $x - p$ lies in the normal cone of the convex set H at p, nearest-point projection satisfies $\pi_H(x) = p$. At a vertex $p = o_k$ we simply have $p \in \partial D_1$ and $\pi_H(p) = p$. Hence $\pi_H(\partial D_1) = \partial H$.

The cyclic word (3) traverses each of the five seam segments A_2A_1, A_1J, JC, JB_1, and B_1B_2 exactly twice, once in each direction. On each outer decagon edge, g is affine and maps that edge onto the corresponding seam segment. Therefore a relative interior point y of any seam segment has preimages p, p' in the relative interiors of two distinct decagon edges. By the first paragraph there are $x, x' \in \partial D_1$ with $\pi_H(x) = p$ and $\pi_H(x') = p'$, so $x \ne x'$ and $F_1(x) = F_1(x') = y$. At the terminal seam vertices A_2, C, and B_2, the two paired boundary arcs meet at the same image vertex; at the intermediate vertices A_1, J, and B_1, the incident paired arcs meet according to the cyclic word (3). In every case the corresponding quotient vertex link, checked in Lemma 4, is a circle rather than an interval. Thus the relative interiors are paired and the endpoints close up without leaving boundary, so the whole disk boundary is self-glued in the image. □

Scaling both the planar domain and the target by R defines F_R and gives $\mathrm{Vol}(B_R) = R^3\,\mathrm{Vol}(B_1)$. Together, Lemmas 3 to 9 prove Theorem 1.

7 Verification protocol and reproducibility

The supplementary archive contains two independent exact verifiers together with a `README.md` giving the exact commands, dependencies, and expected certificate output.

Verifier A

`gyoza_1pleat_exact_certificate.py`.

Target coordinates use Python's `fractions.Fraction`; source metric expressions use exact symbolic arithmetic in $\mathbb{Q}(\sqrt{5})$. Triangle intersections are tested by exact plane-section interval arithmetic.

Verifier B

`gyoza_1pleat_independent_verify.py`.

This implementation does not use SymPy. It contains a hand-written $\mathbb{Q}(\sqrt{5})$ arithmetic type, checks orientability by propagation on the dual graph, and tests triangle intersections by the barycentric feasibility method described in Lemma 5.

Both programs independently return the same exact rational volume and the same intersection counts

$$260 + 130 + 45 = 435.$$

A successful run prints, among other diagnostics,

```
V,E,F = 17 45 30 Euler = 2
triangle pairs = {1: 130, 2: 45, 0: 260}
volume decimal = 0.17984349416291615
all 30 triangle maps are strictly 1-Lipschitz
```

The scale factor $\lambda = 0.99999$ is intentionally conservative. It creates a strict nonexpansion margin while preserving the combinatorial and intersection structure under uniform scaling. The theorem makes no optimality claim about the displayed solid.

8 Discussion

The construction establishes only a lower bound for the maximum volume wrappable by a disk:

$$\sup\{\mathrm{Vol}(B) : B \text{ is wrappable by } D_R\} \geq CR^3.$$

It does not identify the supremum, nor does it claim that one pleat is optimal. More importantly for the interpretation of Nandakumar's conjecture, it does not determine the supremum restricted to convex wrappable bodies. The present theorem should therefore be read as a small explicit and exactly certified witness for the particular nonconvex sphere-comparison problem in Question 1, not as a claim of first abstract existence and not as a proof of Claim 1 for the disk. The Mylar comparison in section 2 illustrates why this distinction matters: the sharp convex body-of-revolution benchmark already improves substantially on the sphere, while our construction improves further and is explicitly nonconvex.

The use of a piecewise-affine source map is also distinct from the problem of finding a neat isometric origami folding. Work of Demaine and Ku shows that, for polygonal boundaries folded nonexpansively at finitely many points, compatible isometric mappings of the polygon interior exist under their model [6]. The present proof does not invoke that theorem: the full non-neat 1-Lipschitz map is written down directly and independently certified for self-intersection.

A natural next problem is to replace the regular decagon by finer polygonal approximations and optimize the crease network while retaining an exact certificate. Another, logically distinct, problem is to understand the best convex body wrappable by a disk. Neither question is needed for the present result: a finite one-pleat construction gives a direct, compact certificate for the specific nonconvex improvement asked for in Nandakumar's Question 1 by a robust margin.

Acknowledgements

The coordinate configuration was found by computer-assisted geometric search. The final proof was reorganized into a finite exact certificate and checked by two independent implementations. External verification of both the mathematical interpretation and the code is encouraged.

References

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